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Master's thesis in Physics and Mathematics

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ABSTRACT

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PREFACE

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Abbreviations test: CNN

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ABBREVIATIONS

BSN body sensor network. v, 5, 7

CNN Convolutional Neural Network. ii, iii, 10–12, 30

ECG electrocardiograms. 1, 6

HAR Human Activity Recognition. iii, v, 1–3, 5–12, 18

MCU microcontroller units. 13

RMS root mean square. vii, 22, 23

TinyML Tiny machine learning. 12

INTRODUCTION

1.1 Introduction

Human Activity Recognition (HAR) has become a central research area in ubiquitous computing due to its wide range of applications in healthcare, rehabilitation, elderly assisted living and human-machine interaction. In particular, wearable sensor-based HAR enables continuous, non-invasive monitoring of physical activity in real-world environments, making it well suited for health monitoring outside clinical settings. This is increasingly important given the aging population and the growing burden of chronic diseases, which place significant pressure on healthcare systems.

Wearable sensors such as accelerometers, gyroscopes, magnetometers, and physiological sensors like electrocardiograms (ECG) are small, low-cost, and energy-efficient, and do not raise the same privacy concerns as vision-based systems. However, relying on a single sensor modality often leads to limited robustness and reduced recognition accuracy, especially in the presence of noise, sensor failure, or ambiguous activities. As a result, multisensor data fusion has emerged as a key strategy to improve reliability, generalization and classification performance in HAR systems.

A typical wearable HAR pipeline consists of sensor data collection, preprocessing and segmentation, feature extraction, classification, and performance evaluation. Sensor data are usually sampled at fixed rates and segmented into short temporal windows, where parameters such as window length and stride influence both recognition performance and system latency. These design choices are particularly important in wearable health monitoring systems, where computational efficiency and energy consumption are critical constraints. Therefore this paper investigates thoroughly how these parameters affect classification accuracy, examines the contribution of different sensor combinations through ablation studies, and analyzes how different sensor combinations compensate for noisy or degraded sensor signals through simulation when making these design choices.

In addition to general activity monitoring, wearable multisensor systems have increasing potential in neurological assessment, i.e. for disorders characterized by abnormal movement patterns such as Parkinson's disease. Parkinsonian tremor is one of the most common motor symptoms and is typically evaluated clinically using visual rating scales, where trained clinicians assign severity scores based

on observed tremor amplitude during specific tasks. Although widely used, such assessments are inherently subjective and provide only short snapshots of symptom severity under controlled clinical conditions.

Wearable sensing offers the possibility of continuous and objective tremor quantification during daily life. Since tremor severity may vary across activities and body parts, combining activity recognition with tremor estimation enables contextualized assessment of symptom expression.

Furthermore, tremor-contaminated motion may alter the sensor patterns on which conventional HAR systems rely, potentially reducing recognition robustness if models are trained only on healthy movement data. This creates a dual motivation, improving HAR robustness under tremor-related signal disturbances, while simultaneously exploiting the same sensor pipeline for tremor severity estimation.

In this work, the HAR framework is therefore extended with an additional tremor classification branch designed to estimate tremor severity on a clinical 0–4 scale. Such a system may support more objective longitudinal monitoring, reduce dependence on repeated in-clinic assessments, and provide new opportunities for relating symptom severity to daily activity patterns over time.

1.1.1 Challenges

Despite their advantages, wearable HAR systems remain sensitive to real-world disturbances such as sensor noise, motion artifacts, and sensor misplacement. For example, sensors worn on top of clothing may experience additional shaking that does not correspond directly to the user’s movement, leading to degraded signal quality. These effects are rarely addressed systematically in the literature, particularly in the context of deep feature-level fusion.

1.1.2 Motivation and Objectives of This Work

The focus of this thesis is feature-level multisensor fusion using deep feature extraction from one-dimensional CNNs applied to wearable sensor time-series data, consisting of accelerometer, gyroscope, magnetometer, and ECG signals. The proposed framework uses separate CNNs for each sensor modality to extract feature embeddings, which are then fused through feature concatenation and classified into one of twelve activity classes.

This work systematically investigates the robustness of deep feature-level fusion under realistic sensor noise and perturbations, including additive white Gaussian noise, signal attenuation, and simulated sensor misplacement. In addition, sensor importance is analyzed through ablation studies and gating mechanisms to understand how different modalities compensate for each other in fused representations. The impact of system-level parameters such as sampling rate, window length, and decision latency is also explored. Finally, autoencoder-based compression of fused feature representations is evaluated to reduce communication and energy consumption in Body Area Networks.

The framework is further extended with a tremor estimation branch to investigate whether feature representations learned for activity recognition can also support tremor severity classification within the same wearable sensing pipeline.

1.2 Goal

The primary goal of this project is to investigate how wearable multisensor HAR systems can be designed to achieve both robustness and efficiency, while also exploring whether the same sensing pipeline can support Parkinsonian tremor severity estimation.

Specifically, the project aims to:

- Evaluate model performance under both clean sensor conditions and simulated sensor degradation, including noise and tremor-like motion artifacts.
- Identify energy- and data-efficient configurations of the HAR system by determining optimal combinations of sampling frequency, window size, and number of sensors, while maintaining acceptable classification performance during degraded sensor conditions.
- Assess whether model robustness to sensor degradation can be improved by incorporating an adaptive fusion mechanism that dynamically estimates sensor quality and adjusts sensor contributions accordingly.
- Investigate whether the same multisensor pipeline can be extended to estimate Parkinsonian tremor severity, enabling simultaneous activity recognition and tremor assessment from wearable sensor data.

Overall, the objective is to understand how sensor configuration, feature fusion strategies, and adaptive robustness mechanisms influence both activity recognition performance and the potential for simultaneous tremor severity assessment in wearable sensing systems.

1.3 TODO

- Sjekk om modeller for lavere seq_len bør kun ha et maxpool1d(2)-lag.
- Finne ut hvor mange runs jeg skal ta per ablation test og per fs test for at det skal være representativt. Så bruke gjennomsnitt av accuracy
- Teste ablation studiene og fs studiene med tremor datasett

I am investigating if the optimal sensor (and frequency) combination changes when there are more corruption and degradation on the sensor data. I will explore what is the optimal combination in the degraded scenario, and investigate if adaptive gating weights will improve the accuracy.

1.3.1 Motivation

- increased healthcare needs due to aging population and chronic health conditions
- monitoring of patience

The research investigates if the state of the art HAR ANN models are robust enough to handle corrupted datam with artifacts such as electronical noise, magnetic field disturbances and tremor. This will provide valuable insight in relation

to the development of HAR systems for everyday use and for patients suffering from tremor. It will provide results that show if the models need to be trained on specialized data, or if models trained on clean data are robust enough to handle corrupted data as well. It will also give insights on how well one model can handle both clean and corrupted data, and thereby insinuate if it is enough to provide one model or if there is a need for multiple, specialized models for different users.

1.4 Project description

1.4.1 Stakeholders

BACKGROUND

2.1 Body Sensor Networks

A body sensor network (BSN) consists of multiple wearable or body-mounted sensor nodes that wirelessly collect physiological and inertial signals from the human body. The sensor nodes may be worn on the body surface, placed close to the body or be implanted, and are typically connected through a wireless communication network [1].

BSNs commonly include multiple wearable sensors which together capture information related to body movement, orientation and physiological state. The collected sensor data can then be processed using signal processing and machine learning techniques to extract meaningful information [1].

In many wearable systems, the sensor nodes perform some degree of local data processing before transmitting data to a central coordinator device, such as a smartphone, smartwatch or embedded processing unit. This architecture enables continuous monitoring while supporting mobility and real-time analysis.

BSNs are widely used in applications such as health monitoring, rehabilitation, sports analysis and HAR [2], where multiple wearable sensors are combined to improve robustness and recognition performance.

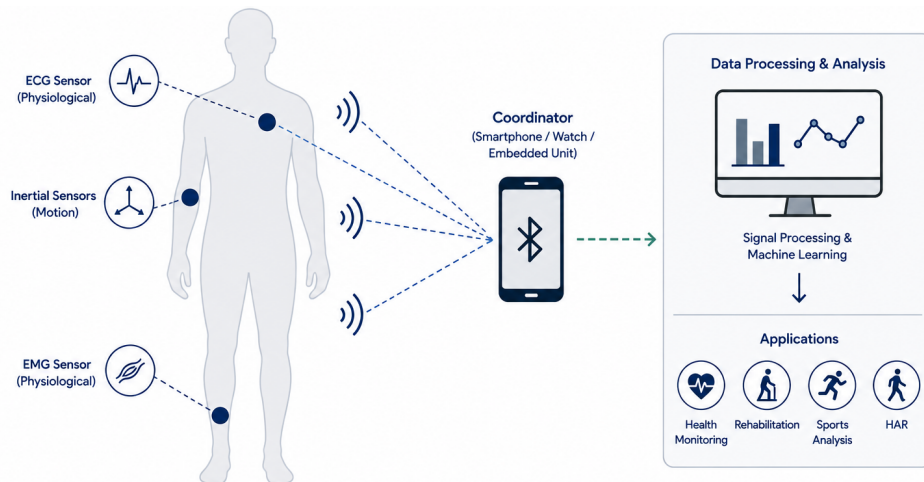


Figure 2.1.1: BSN (AI generated)

2.1.1 Wearable Sensors in HAR systems

Wearable HAR systems commonly use inertial sensors, including accelerometers, gyroscopes and magnetometers. Accelerometers measure linear acceleration and are widely used due to their low cost and low power consumption [3], but they are sensitive to sensor placement, noise and gravity effects [4]. Gyroscopes provide angular velocity information that helps distinguish rotational movements but suffer from drift over time [4]. Magnetometers measure the Earth’s magnetic field and can assist with orientation estimation, though they are sensitive to magnetic disturbances [4]. Fusing these inertial sensors can compensate for individual weaknesses and provide more stable motion representations. In particular, multisensor fusion can help reduce the effect of sensor drift, improve orientation estimation, support recognition of transitional activities and enable more accurate posture identification [4].

In addition to inertial sensors, ECG signals have been widely used for physiological monitoring. Although ECG is primarily intended for cardiac analysis, both heart rate variability and motion artifacts induced by body movements contain information related to physical activity. Prior studies have shown that these ECG-derived features can contribute to activity recognition [5]. Furthermore, multi-lead ECG systems provide complementary information across leads, enabling improved recognition performance compared to single-lead ECG [5].

2.1.2 Motivation for Multi-Sensor Data Fusion

Although wearable HAR systems can operate using a single sensor modality, single-sensor systems often suffer from several limitations. Individual sensors may be affected by noise, uncertainty, sensor placement variations and temporary sensor failures. In addition, a single wearable sensor only captures information from a limited body location, which may reduce the ability to accurately characterize complex activities [1].

Multi-sensor data fusion addresses these limitations by combining information from multiple sensor sources. By integrating complementary or redundant information, fusion methods can reduce the effects of noise and uncertainty while improving robustness and reliability. Using multiple sensor streams can reduce the influence of noisy measurements, while redundant sensing increases fault tolerance in the event of sensor failure. In addition, combining information from multiple sensors and body locations can improve robustness against environmental interference by increasing the dimensionality of the measurement space and providing more stable activity representations. Fusion can also increase confidence in the estimated activity by exploiting information from multiple body locations and sensor modalities simultaneously [1].

Another important advantage of multisensor fusion is that different sensor modalities capture different aspects of human activity. For example, accelerometers mainly capture translational motion, gyroscopes provide rotational information, while physiological sensors such as ECG can provide complementary information related to physical exertion and body state. Combining these sources can therefore provide richer and more discriminative representations of human activities.

For these reasons, multi-sensor fusion has become an important component in modern wearable HAR systems.

2.2 Multisensor Data Fusion in HAR

Multisensor data fusion aims to combine information from multiple sensor modalities in order to reduce uncertainty, improve robustness and increase classification performance. Fusion methods can be categorized according to both the relationship between the sensor sources [6] and the abstraction level at which the fusion is performed [luo1989].

2.2.1 Fusion relationships

Based on the relationship between the sensor inputs, fusion methods are commonly categorized as *complementary*, *redundant/competitive*, or *cooperative* fusion [6].

1. *Complementary fusion:*

In complementary fusion, different sensors capture different aspects of the same activity or phenomenon. Combining these complementary perspectives can provide a more complete representation than any individual sensor alone and is commonly used to improve system accuracy and reliability [1].

For example, accelerometers mainly capture translational movement, while gyroscopes capture rotational motion. Combining these modalities can therefore improve characterization of complex human activities.

2. *Redundant/competitive fusion:*

Competitive fusion combines overlapping information from multiple equivalent sensors observing similar phenomena. Since several sensors provide related information, the fused representation becomes more robust to noisy measurements, sensor failures and uncertainty.

This is uncommon in BSN systems, as sensors are often placed on different body locations and therefore capture different aspects of the movement, providing complementary rather than fully redundant information [1].

3. *Cooperative fusion:*

Cooperative fusion occurs when information from multiple sensors is combined to generate new information that would not be obtainable from the individual sensors independently.

In wearable sensing systems, combining motion information from multiple body locations may enable detection of coordinated movement patterns or transitional activities that cannot be reliably identified using only a single sensor. Cooperative fusion is therefore common in multisensor HAR systems [1].

2.2.2 Fusion levels

In HAR, fusion methods are commonly categorized into three levels: *data-level fusion*, *feature-level fusion*, and *decision-level fusion* [7].

1. *Data-level fusion*:

Data-level fusion, also referred to as signal-level fusion, represents the lowest level of abstraction and combines raw sensor signals directly before feature extraction or classification. Since fusion is performed on the original sensor measurements, no information is lost during feature extraction or processing. However, corrupted or noisy sensor signals may directly affect the fused representation and propagate through the subsequent processing stages [1, 2].

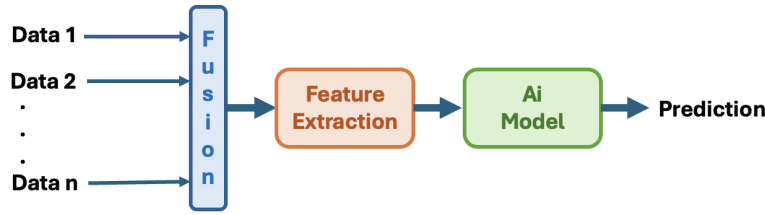


Figure 2.2.1: Data-level fusion

2. *Feature-level fusion*:

Feature-level fusion combines features extracted independently from the different sensor modalities. Since the fusion is performed after feature extraction, each sensor stream can first be processed individually before being merged into a combined representation. This provides greater flexibility compared to data-level fusion and enables integration of complementary information from multiple sensors [1, 2].

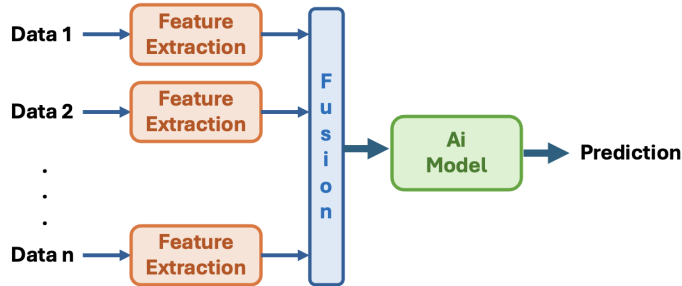


Figure 2.2.2: Feature-level fusion

3. *Decision-level fusion*:

Decision-level fusion performs fusion at the output stage by combining decisions or predictions from multiple sensor streams or classifiers. Common

approaches include majority voting and weighted voting. By combining multiple independent decisions, decision-level fusion can improve the reliability of the final prediction [1, 2].

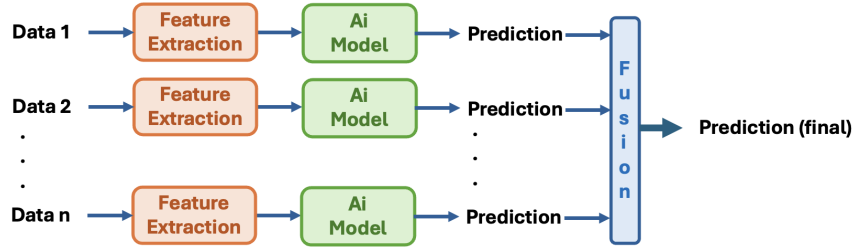


Figure 2.2.3: Decision-level fusion

2.3 Feature extraction

In HAR systems, feature extraction aims to transform raw sensor signals into informative representations that can be used for activity classification. This is commonly achieved either through handcrafted feature extraction or through automatically learned representations using deep learning methods.

2.3.1 Handcrafted feature extraction

In handcrafted feature extraction, features are manually computed from the pre-processed sensor signals and combined into a feature vector, which is then used as input to a machine learning classifier [8].

Handcrafted features commonly include simple statistical measures such as mean, median, variance, standard deviation and RMS values, as well as more complex time- and frequency-domain features such as signal magnitude area, power spectral density and Fourier transform coefficients [4].

Handcrafted features are often computationally efficient, making them suitable for systems with limited computational resources or smaller datasets. However, designing handcrafted features requires significant manual effort and expert knowledge, and the selected features may not generalize well across different datasets, sensor modalities or deployment conditions [8].

To address these limitations, deep learning-based feature extraction has become increasingly popular in wearable HAR systems. Instead of relying on manually engineered signal descriptors, deep learning models learn feature representations directly from the raw sensor data through hierarchical neural network architectures.

2.3.2 Deep learning feature extraction

Deep learning has been increasingly popular in HAR as it reduces the reliance on handcrafted features and expert domain knowledge. It enables the model to automatically learn feature representations directly from raw sensor data, reducing

the need for manually engineered features and domain-specific feature design [4]. Instead of relying on predefined features, deep learning models learn hierarchical feature representations through multiple layers of neural networks.

In wearable HAR systems, deep learning methods have become increasingly popular due to their ability to model complex and nonlinear relationships in multimodal sensor data. Compared to handcrafted approaches, deep learning models can learn more discriminative and robust feature representations while reducing the manual effort associated with feature engineering [8, 4].

Unlike traditional handcrafted feature fusion, deep learning-based feature extraction enables each sensor modality to be mapped into its own latent representation space. These learned feature embeddings can subsequently be combined through feature-level fusion and jointly classified, even when the underlying sensor streams differ in sampling frequency, pre-processing strategy or signal characteristics. This provides greater flexibility and scalability for multisensor integration compared to conventional handcrafted fusion approaches.

Several deep learning architectures have been applied to HAR, including autoencoders, recurrent neural networks (RNNs), long short-term memory (LSTM) networks, and Convolutional Neural Network (CNN)s [4]. Autoencoders are commonly used for unsupervised feature learning and dimensionality reduction, while recurrent architectures such as RNNs and LSTMs are particularly useful for modeling temporal dependencies in sequential sensor signals [4].

CNN-based feature extraction has become especially common in wearable HAR systems, as CNNs are capable of automatically learning locally correlated and translationally robust feature representations directly from raw sensor signals [4].

Through convolution and pooling operations, CNNs can efficiently capture local temporal patterns and hierarchical representations in sensor time-series data [4]. In addition, CNNs are computationally efficient compared to many recurrent architectures and are well suited for parallel processing, making them attractive for resource-constrained wearable sensing systems [4].

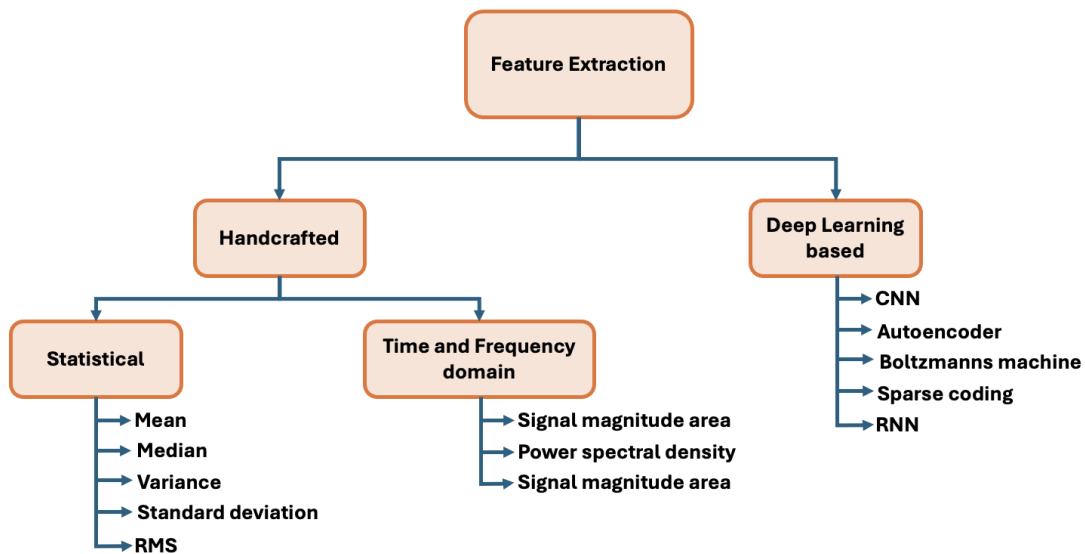


Figure 2.3.1: Overview of common handcrafted and deep learning-based feature extraction approaches used in wearable HAR systems [8].

2.4 Convolutional Neural Networks (CNN)

Convolutional Neural Networks (CNNs) are among the most widely used deep learning architectures for wearable HAR systems due to their ability to automatically learn discriminative feature representations directly from raw sensor signals [4]. Originally developed for image analysis, CNNs have also proven highly effective for time-series sensor data by learning local temporal patterns and hierarchical feature representations [9].

A CNN typically consists of convolutional, activation, pooling and fully connected layers, often combined with regularization techniques such as dropout. The convolutional layers apply learnable filters, referred to as kernels, across the input signal in order to generate feature maps that capture local patterns and correlations within the data. During training, the network automatically learns filter parameters that are useful for distinguishing between different classes.

For wearable sensor signals, one-dimensional convolutions can be used, where the convolutional kernels slide along the temporal dimension of the signal. This allows the network to learn local temporal dependencies and motion-related patterns directly from the sensor time series.

After each convolutional operation, batch normalization layers are commonly used to normalize intermediate feature distributions during training, which can improve training stability and convergence speed. Nonlinear activation functions such as ReLU are then applied to introduce nonlinearity and enable the network to learn more complex patterns.

Pooling layers are commonly inserted between convolutional layers to reduce the dimensionality of the feature maps and decrease the computational complexity of the network. Pooling also provides increased robustness to small temporal shifts and signal distortions by summarizing local regions of the feature maps. Common pooling operations include max pooling and average pooling.

Dropout layers are also commonly used to reduce overfitting by randomly deactivating neurons during training.

After several stages of convolution and pooling, the extracted feature maps are flattened and passed to fully connected layers, which perform the final classification task. The fully connected layers combine the learned feature representations and estimate the probability of each activity class, using activation functions such as softmax.

CNNs are generally computationally efficient and well suited for parallel processing, making them attractive for resource-constrained wearable sensing systems [4]. CNNs have also demonstrated strong performance in wearable HAR due to their ability to automatically learn robust feature representations without requiring manually engineered features.

In multisensor HAR systems, CNNs can either be applied directly to fused sensor signals or independently to each sensor modality before feature-level fusion. In the latter approach, each sensor stream is mapped into its own latent feature representation space, after which the learned embeddings can be combined and jointly classified.

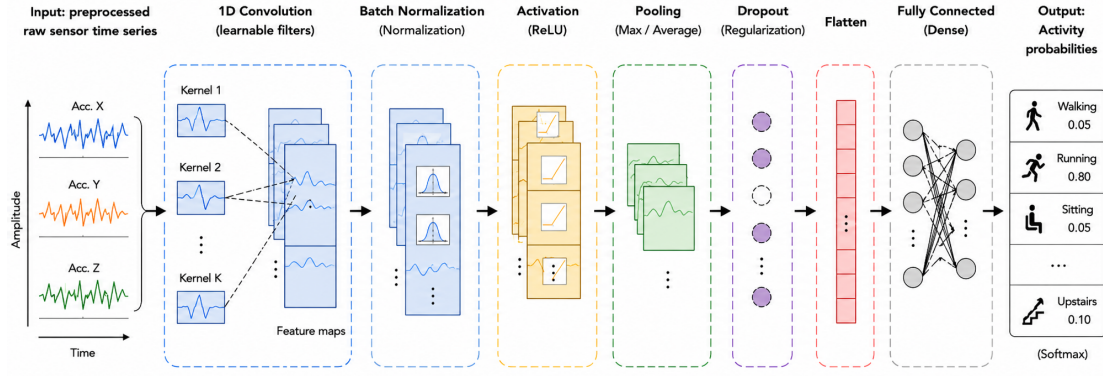


Figure 2.4.1: Illustration of a one-dimensional CNN pipeline for wearable HAR. Raw sensor time-series signals are processed through convolutional, normalization, activation, pooling and fully connected layers in order to learn hierarchical feature representations and perform activity classification. (AI generated figure).

2.4.1 CNNs for deep feature extraction

In addition to performing classification, CNNs can also be used as automatic feature extractors. Through convolution and pooling operations, the network learns highly informative representations of the input sensor signals.

The extracted features are typically obtained from the intermediate layers of the network before the final classification stage. After the convolutional and pooling layers, the resulting feature maps are flattened into a compact feature vector, often referred to as an embedding or latent representation.

These embeddings contain high-level information about the sensor signals and can subsequently be used as input to downstream classifiers or feature-level fusion methods in multisensor HAR systems.

Figure 2.4.2 illustrates how feature embeddings can be extracted from a CNN before the final classification stage.

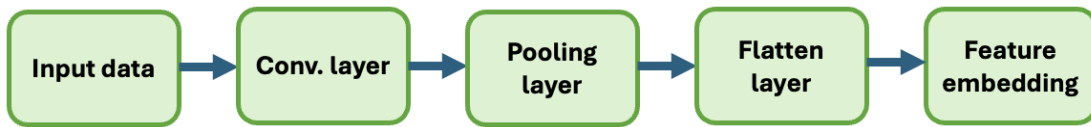


Figure 2.4.2: CNN-based feature extraction, where feature embeddings are extracted from intermediate network layers before classification.

2.5 TinyML and resource-constrained machine learning

Tiny machine learning (TinyML) refers to machine learning systems designed for highly resource-constrained edge devices, such as microcontrollers and low-power embedded systems. The objective of TinyML is to enable on-device inference, meaning that trained machine learning models can perform predictions directly on

low-power embedded devices while having low energy consumption and acceptable inference accuracy [10].

TinyML has become increasingly relevant for wearable sensing applications, including healthcare monitoring, activity recognition and always-on wake modules [10]. In such systems, the available computational resources are often highly limited, motivating the need for compact and computationally efficient machine learning models.

Typical TinyML systems are based on low-power microcontroller units (MCU)s with strict hardware constraints. Prior work describes representative TinyML platforms as having less than approximately 512 KB SRAM, less than 2 MB Flash memory, computational capabilities below 1 GOP/s, and power budgets below 1 W [11].

Based on these representative TinyML constraints, the target system in this work is designed with the following resource considerations in mind:

1. *The total model storage should remain below approximately 2 MB Flash memory. Assuming 32-bit floating point parameters, this corresponds to a maximum parameter budget of approximately 524 000 trainable parameters.*
2. *Runtime memory usage, including intermediate feature maps, embeddings and inference buffers, should remain compatible with systems providing less than approximately 512 KB SRAM. This motivates compact convolutional feature maps and lightweight embedding representations.*
3. *The computational complexity of the network should remain compatible with systems providing less than approximately 1 GOP/s computational capability. This motivates compact CNN architectures, reduced sensor configurations and lower sampling frequencies.*
4. *The framework should support energy-efficient operation suitable for battery-powered wearable and embedded edge devices. This motivates minimizing sensor count, sampling frequency and data movement during inference.*

Compact architectural design is therefore essential for TinyML systems. For example, Xu et al. proposed the EtinyNet architecture for TinyML applications, achieving competitive classification accuracy with only 477 KB of model weights [11].

Another important aspect of TinyML is reducing the amount of input data processed by the system. In wearable sensing applications, energy consumption is not only affected by the neural network itself, but also by the number of active sensors, the sampling frequency of each sensor and the amount of data transferred and processed. Reducing the amount of sensors and their sampling frequency can therefore significantly decrease both computational load and energy consumption, creating an important trade-off between recognition performance and resource efficiency.

METHODOLOGY

3.1 Dataset

The research in this work are based on the publicly available MHEALTH dataset [12], which contains multimodal wearable sensor recordings collected during a range of physical activities performed by healthy subjects. The dataset includes inertial measurements from accelerometers, gyroscopes and magnetometers placed at different body locations, together with ECG recordings.

The MHEALTH dataset includes wearable sensors placed at three body locations: the chest, left wrist and right ankle. The left wrist and right ankle sensor nodes each contain a tri-axial accelerometer, gyroscope and magnetometer, while the chest sensor node contains a tri-axial accelerometer together with an ECG sensor. Figure 3.1.1 illustrates the sensor placements and corresponding sensor modalities used in the dataset.

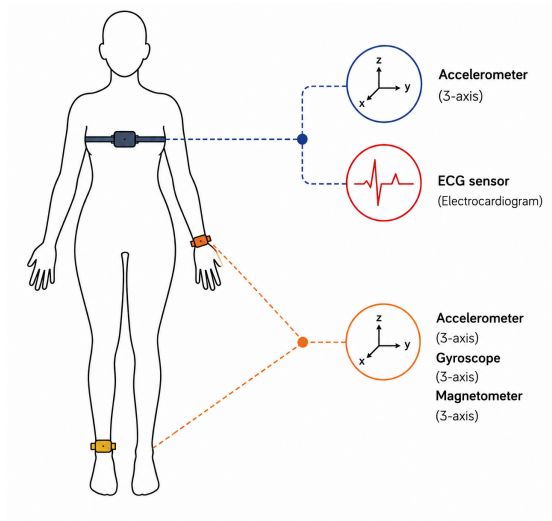


Figure 3.1.1: sensor placements (AI generated figure).

The dataset contains recordings from 10 subjects performing 12 physical activities, including both stationary and dynamic movements:

1. *Standing still*

2. *Sitting and relaxing*
3. *Lying down*
4. *Walking*
5. *Climbing stairs*
6. *Waist bends forward*
7. *Frontal elevation of arms*
8. *Knees bending*
9. *Cycling*
10. *Jogging*
11. *Running*
12. *Jump front and back*

The raw sensor signals are sampled at 50 Hz.

3.1.1 Dataset distribution

The activity labels L1–L12 correspond directly to the numbered activities listed in Section 3.1. Overall, the dataset is relatively balanced across activities, with most activities containing approximately 30 000 samples each. However, activity L12 (*Jump front and back*) contains significantly fewer samples, with approximately 10 342 samples in total. The distributions of samples across the activity classes are illustrated in Figure 3.1.2.

Activities L1–L5 and L9–L11 contain identical numbers of samples, while moderate variations are observed for activities L6–L8. These differences are caused by variations in recording duration.

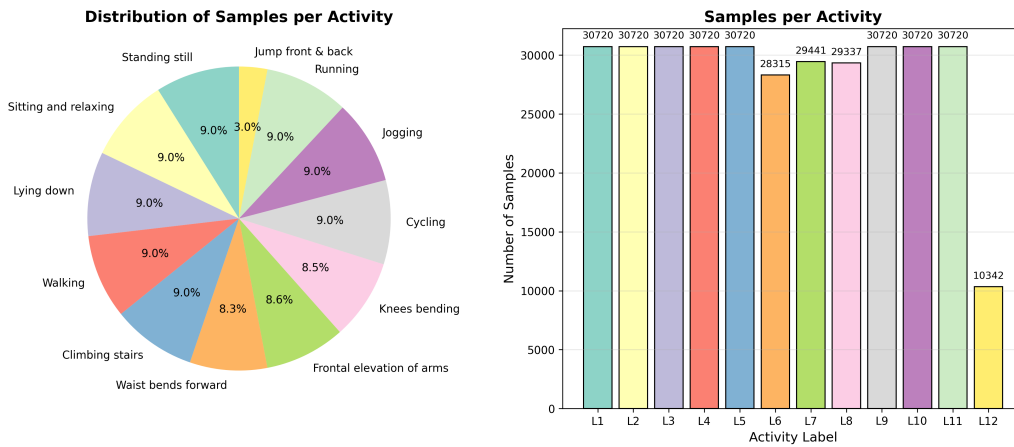


Figure 3.1.2: Distribution of samples across the activity classes. The left pie chart shows the relative class distribution, while the right bar chart shows the total number of samples for each activity label.

The subject distribution is also relatively balanced, with the total number of samples per subject ranging from 32 205 to 35 532 samples. Most of the variation between subjects originates from activities L6–L8, while smaller differences are observed for activity L12. Figure 3.1.3 visualizes the number of samples contributed by each subject and activity.

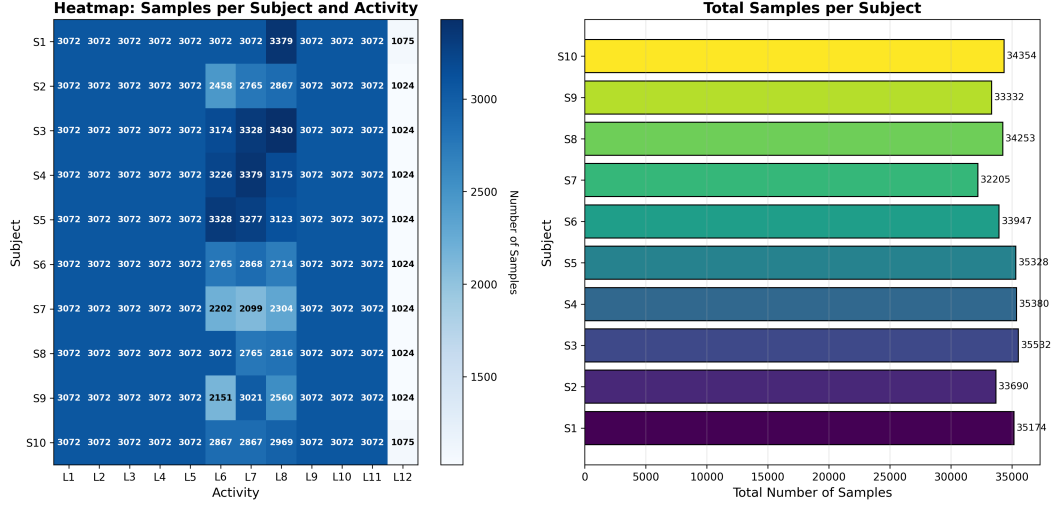


Figure 3.1.3: Distribution of samples across subjects and activities in the MHEALTH dataset. The heatmap illustrates the number of samples for each subject–activity pair, while the bar plot shows the total number of samples contributed by each subject.

3.2 Data Preprocessing

All raw sensor signals from the MHEALTH dataset are preprocessed using a common pipeline before being used for model training and evaluation. The preprocessing pipeline consists of the following steps, executed in order: *segmentation*, *resampling*, and *per-window normalisation*. All preprocessing operations are applied independently to each sensor channel.

3.2.1 Segmentation

The continuous recordings are divided into fixed-length windows using a sliding-window approach. A window length of 4 s and a stride of 4 s are used, resulting in non-overlapping windows. At the original sampling rate of 50 Hz, each window contains 200 samples. Segmentation is applied independently within each activity recording for each subject, so that no window spans two different activity labels.

The choice of a 4 s window reflects a trade-off between temporal context, the number of available training samples, and system latency. A shorter window increases the number of samples and reduces the delay from event onset to recognition, but provides less temporal context to the model. A longer window captures richer temporal structure at the cost of fewer training samples per activity and higher end-to-end latency, since a full window must be collected before processing can begin.

The non-overlapping stride (stride = window length) avoids introducing correlation between adjacent windows, which helps maintain unbiased subject-based train/validation/test splits while also reducing the risk of overfitting.

3.2.2 Resampling

Following segmentation, each window may optionally be resampled to a target sampling rate f_s using band-limited interpolation (`scipy.signal.resample`). Resampling is applied channel by channel. The step is included in the pipeline to support experiments at reduced sampling rates without changing the upstream segmentation logic.

3.2.3 Z-score Normalization

After resampling, each window is normalised on a per-channel basis using z-score normalization:

$$\hat{x}_c[n] = \frac{x_c[n] - \mu_c}{\sigma_c}, \quad (3.1)$$

where μ_c and σ_c are the mean and standard deviation of channel c computed over the L samples within the window. A lower bound of 10^{-8} is applied to σ_c to handle near-constant channels. Z-score normalization transforms each channel to have zero mean and standard deviation of one within the window. Normalizing within each window rather than across the full recording makes the representation invariant to long-term drift and absolute signal amplitude, while preserving the relative temporal dynamics within the window.

3.2.4 Complete Preprocessing Pipeline

Figure 3.2.1 summarizes the complete preprocessing pipeline applied to the raw sensor signals before being used as input to the HAR model.

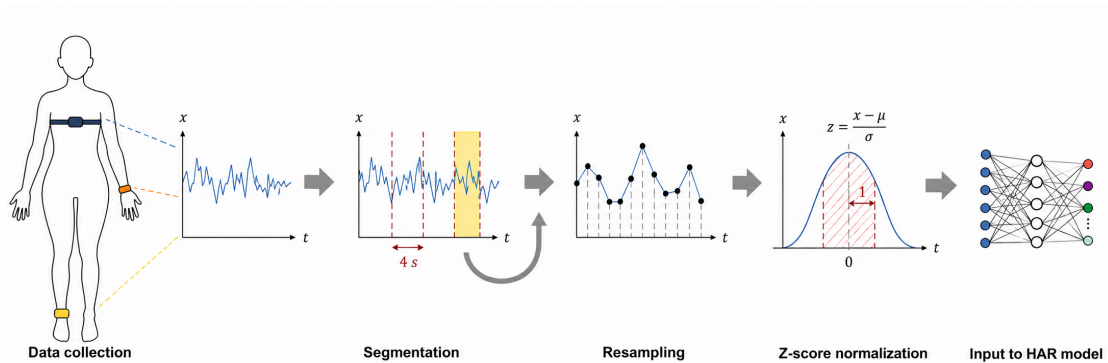


Figure 3.2.1: Overview of the preprocessing pipeline applied to the raw sensor signals. The pipeline consists of segmentation into non-overlapping windows, optional resampling to a target sampling frequency and per-window z-score normalization before the signals are used as input to the HAR model. (AI generated figure)

3.3 Dataset splits

- subject based

3.4 Signal corruption framework

To evaluate the robustness of wearable HAR systems under realistic sensor disturbances, several signal corruption scenarios are simulated and injected directly into the raw sensor signals prior to preprocessing.

The corruption framework is designed to emulate common degradation sources in wearable sensing systems, including motion artifacts, electronic sensor noise, signal attenuation, temporary signal loss and tremor-related perturbations. All corruptions are applied before preprocessing in order to preserve realistic signal degradation throughout the preprocessing pipeline. The processing flow is:

Raw signal $\rightarrow$ Noise injection $\rightarrow$ Resampling $\rightarrow$ Per-window z-score $\rightarrow$
 $\rightarrow$ Feature extraction and fusion

3.4.1 Additive white Gaussian noise (AWGN)

Sensor noise was simulated using additive white Gaussian noise applied independently to each sensor channel:

$$x_{\text{noisy}} = x + n$$

where:

$$n \sim \mathcal{N}(0, \sigma^2)$$

The noise standard deviation was scaled relative to the RMS amplitude of the corresponding sensor window:

$$\sigma = \alpha \cdot \text{RMS}(x)$$

where α controls the corruption strength.

This corruption models common electronic and environmental noise sources in wearable sensing systems, including thermal sensor noise, electromagnetic interference, quantization effects and motion-related signal disturbances.

3.4.2 Orientation perturbation

To simulate sensor misalignment and variations in wearable placement, random three-dimensional rotation perturbations were applied to the tri-axial IMU signals. For each affected window, the original sensor measurements were multiplied by a random rotation matrix:

$$\mathbf{x}'(t) = R\mathbf{x}(t), \tag{3.2}$$

where $\mathbf{x}(t) \in \mathbb{R}^3$ denotes the original sensor vector and $R \in \mathbb{R}^{3 \times 3}$ is a randomly sampled rotation matrix.

This corruption simulates realistic changes in sensor orientation that may occur during wearable deployment while preserving the physical structure of the underlying motion. Rotation angles up to 45° were used in order to simulate severe sensor misplacement scenarios.

For each IMU sensor node, the same rotation matrix was applied consistently across the accelerometer, gyroscope and magnetometer channels within the same window.

3.4.3 Partial signal loss

Temporary signal degradation and communication failures were simulated by randomly masking contiguous temporal segments corresponding to 10% of each sensor window with zeros.

This corruption models temporary signal degradation scenarios such as packet loss, wireless communication instability and intermittent sensor disconnections.

3.4.4 Full signal loss

Complete sensor failure was simulated by replacing all samples within a sensor window with zeros,

$$y(t) = 0.$$

This corruption represents severe degradation scenarios in which the sensor provides no usable information, such as hardware failure, battery depletion or complete communication loss.

3.5 Parkinsonian tremor simulation

Tremor was simulated using a second-order nonlinear stochastic dynamical model, inspired by established mathematical models of pathological tremor [13, 14]. Such models capture key characteristics of tremor signals, including non-sinusoidal oscillations, amplitude variability, and stochastic fluctuations, which are not adequately represented by purely linear or sinusoidal processes.

The tremor process was modeled as a stochastic van der Pol oscillator,

$$\ddot{x}(t) - \mu(1 - x(t)^2)\dot{x}(t) + x(t) = \sigma\xi(t), \quad (3.3)$$

where $x(t)$ denotes the tremor state, μ controls the nonlinearity of the oscillation, σ determines the stochastic excitation strength, and $\xi(t)$ is zero-mean Gaussian white noise. The system was numerically integrated using the Euler–Maruyama method and resampled to match the sensor sampling frequency.

To account for differences between inertial sensor modalities, the position-like state $x(t)$ was used to perturb the accelerometer and magnetometer signals, while the velocity-like state $\dot{x}(t)$ was used to perturb the gyroscope signals, reflecting their respective sensitivity to translational and rotational motion.

Accordingly, the scalar tremor signal $s(t)$ was defined as

$$s(t) = \begin{cases} x(t), & \text{for accelerometer and magnetometer,} \\ \dot{x}(t), & \text{for gyroscope.} \end{cases}$$

The scalar tremor signal was projected onto a fixed three-dimensional unit vector, $\mathbf{v}$, and added to the original sensor measurements, $\mathbf{z}(t)$,

$$\mathbf{y}(t) = \mathbf{z}(t) + s(t) \mathbf{v}, \quad (3.4)$$

ensuring physically consistent perturbations across sensor axes and modalities.

The tremor amplitude was controlled by scaling the simulated signal to pre-defined RMS levels for each modality, thereby adjusting tremor severity without altering the underlying activity labels.

The final tremor amplitude was scaled according to both activity type and sensor location,

$$A_{\text{scaled}} = A \cdot \beta_{\text{activity}} \cdot \gamma_{\text{location}}, \quad (3.5)$$

where A is a base tremor amplitude, β_{activity} is an activity-dependent scaling factor and γ_{location} is a body-location-dependent scaling factor.

This formulation ensures that all sensors within the same temporal window remain linked to the same underlying tremor severity while still reflecting physiologically realistic differences between activities and sensor locations.

3.5.1 Initial literature-informed parameter selection

The initial tremor simulation parameters were selected based on physiological tremor literature, reported clinical observations and previously reported characteristics of Parkinsonian tremor.

Frequency

Each signal window was simulated with a tremor frequency randomly sampled in the range 3.5–7 Hz. Parkinsonian tremor is most commonly reported in the range of approximately 4–6 Hz, although broader ranges of about 3–7 Hz have been described in the literature [deutschl2000, elble1990]. Accordingly, a slightly extended range of 3.5–7 Hz was used in this work.

Base Tremor Amplitude (A)

The tremor amplitude model was designed to represent clinically realistic Parkinsonian tremor severities corresponding to the Unified Parkinson’s Disease Rating Scale (UPDRS).

To obtain physiologically realistic amplitude ranges, RMS acceleration and angular velocity values were derived from wrist-mounted IMU measurements reported by Keba et al. [15] across clinical Parkinsonian tremor severity scores using the UPDRS.

Table 3.5.1: Approximate root mean square (RMS) ranges extracted from wrist tremor boxplots [15].

Score	RMS_{acc} (m/s ²)	RMS_{gyro} (deg/s)
1	0.046–0.10	0.42–1.11
2	0.246–0.497	4.8–11
3	1.10–1.96	17–25
4	4.65–5.10	62–64

Approximate median values were visually extracted from boxplots reporting RMS acceleration (cm/s²) and RMS angular velocity (deg/s) for different clinical tremor severity scores. The resulting approximate ranges are summarized in Table 3.5.1.

To preserve a consistent relationship between accelerometer and gyroscope tremor amplitudes in the simulation, the gyroscope RMS was modeled as

$$RMS_{gyro} = k_g \cdot RMS_{acc}, \quad (3.6)$$

where RMS_{acc} is measured in m/s² and RMS_{gyro} in deg/s, giving k_g the unit deg/s per m/s².

Midpoint values were used to compute k_g for each severity level:

$$\textbf{Score 1: } RMS_{acc} \approx 0.073, \quad RMS_{gyro} \approx 0.765$$

$$k_g = \frac{0.765}{0.073} \approx 10.5$$

$$\textbf{Score 2: } RMS_{acc} \approx 0.372, \quad RMS_{gyro} \approx 7.9$$

$$k_g = \frac{7.9}{0.372} \approx 21.3$$

$$\textbf{Score 3: } RMS_{acc} \approx 1.53, \quad RMS_{gyro} \approx 21$$

$$k_g = \frac{21}{1.53} \approx 13.7$$

$$\textbf{Score 4: } RMS_{acc} \approx 4.88, \quad RMS_{gyro} \approx 63$$

$$k_g = \frac{63}{4.88} \approx 12.9$$

The resulting estimates are summarized in Table 3.5.2.

Table 3.5.2: Estimated k_g values per clinical tremor score based on RMS values.

Score	k_g
1	10.5
2	21.3
3	13.7
4	12.9

Because the values were visually extracted from boxplots and may be affected by sampling variability and logarithmic scaling, the raw k_g estimates were slightly smoothed to obtain more robust and stable modeling parameters.

The final values used in the simulation are shown in Table 3.5.3.

Table 3.5.3: Severity intervals used in simulation, aligned with empirically observed wrist RMS ranges.

Score	k_g	RMS_{acc} interval (m/s ²)
1 (mild)	11	0.05–0.10
2 (mild–moderate)	22	0.25–0.5
3 (moderate–severe)	14	1.10–2.0
4 (severe)	13	4.65–5.10

For each signal window, a tremor severity score was first sampled and an accelerometer RMS value was randomly selected within the corresponding severity-specific interval shown in Table 3.5.3. This value was assigned to the wrist accelerometer signal, which served as the primary amplitude reference for the simulated tremor.

The corresponding gyroscope RMS amplitude was derived using the severity-dependent scaling factor k_g .

For magnetometer signals, tremor was modeled indirectly through tremor-induced rotational perturbations derived from the gyroscope tremor component. Incremental rotational perturbations were accumulated over time and applied to the original magnetometer vectors, producing variations consistent with orientation-dependent magnetic field measurements.

Activity-Dependent Modulation (β)

Clinical evidence indicates that postural tremor amplitude is lower than resting tremor in approximately 70% of Parkinson’s disease patients, equal in 25%, and higher in only 5% [16]. Accordingly, tremor amplitude was modeled as highest during rest and progressively reduced during voluntary and high-intensity movements.

Since A was selected from studies reporting hand rest tremor amplitudes, the activity modulation was normalized such that $\beta = 1.0$ corresponds to baseline resting tremor (i.e., sitting/relaxing and quiet standing). Values $\beta < 1$ were assigned to voluntary and high-intensity movements to model the commonly observed reduction of tremor amplitude during action [16].

The activity-dependent modulation factor β was defined as shown in Table 3.5.4. Values were chosen to reflect gradual tremor suppression with increasing motor activation.

Table 3.5.4: Activity-dependent modulation factor (β), normalized such that $\beta = 1.0$ corresponds to baseline resting tremor amplitude.

Activity	β	Rationale
Sitting / Relaxing	1.00	resting tremor
cStanding still	1.00	resting tremor
Lying down	1.00	resting tremor
Walking	0.90	Partial tremor suppression
Waist bends	0.90	partial tremor suppression
Arm elevation	0.80	partial tremor suppression
Knees bending	0.85	partial tremor suppression
Cycling	0.80	partial tremor suppression
Climbing stairs	0.75	high intensity, higher tremor suppression
Jogging	0.70	high intensity, higher tremor suppression
Running	0.65	high intensity, higher tremor suppression
Jumping	0.70	high intensity, higher tremor suppression

Body-Location Scaling (γ_{location})

The base tremor amplitude A was derived from wrist-mounted IMU measurements reported by Keba et al. [15], and therefore represents upper-limb rest tremor amplitude.

Parkinsonian tremor is most commonly observed in the upper extremities, while lower-limb and axial tremor are generally less pronounced [**practicalneurology_tremor**, **schwingenschuh2025tremor**]. Accordingly, tremor amplitudes for other body locations were scaled relative to the arm tremor amplitude.

The initial body-location scaling factors were defined as

$$\gamma_{\text{location}} = \begin{cases} 1.0, & \text{arm} \\ 0.8, & \text{ankle} \\ 0.0, & \text{chest} \end{cases}$$

where arm sensors were assumed to exhibit the strongest tremor amplitudes, ankle sensors moderate tremor amplitudes, and no tremor was simulated for the chest sensor in the initial model.

These initial scaling factors were selected based on typical clinical tremor distribution patterns reported in the literature and were later refined through empirical analysis of Parkinsonian IMU data.

3.5.2 Empirical analysis and refinement of the tremor simulation model

To ensure that the simulated tremor model reflects realistic characteristics of Parkinsonian tremor, an empirical analysis was conducted using real patient data. This was done by comparing signal characteristics with those observed in real patient data and by evaluating how well a model trained on simulated data generalizes to real Parkinson activity data.

3.5.3 Dataset for tremor analysis

The dataset provided by Russell et al. (2023) [17], referred to as *dataset 2*, was used for this analysis. The dataset contains kinematic data collected from 15 patients diagnosed with Parkinson’s disease and 19 healthy control participants.

Each subject was equipped with five 9-axis IMUs placed on the upper right arm, upper left arm, lower right arm, lower left arm and on the head. The participants performed a calibration pose (standing with their arms at their sides) followed by three activities of daily living: making toast, putting on a cardigan, and unlocking and opening a door. The IMUs recorded time-series data from accelerometers, gyroscopes and magnetometers throughout each activity.

The dataset does not include clinical tremor severity annotations. Therefore, Parkinson patient data was treated as a *tremor present* dataset without ground truth severity labels.

Feature extraction

For each window, a set of features was computed to characterize both overall motion and tremor-specific behavior:

- RMS amplitude of accelerometer signals
- RMS amplitude of gyroscope signals
- Tremor-specific RMS amplitude computed within the tremor frequency band (3–7 Hz)
- Dominant frequency of the signal
- Ratio between tremor RMS and total RMS
- Ratio between gyroscope and accelerometer RMS (k_g)

Aggregation levels

The extracted features were analyzed across multiple aggregation levels to capture different aspects of tremor behavior:

- *Global statistics:* Overall distributions of amplitude, frequency and tremor prevalence across the entire dataset.
- *Activity-wise analysis:* Feature distributions were computed separately for each activity, allowing assessment of how tremor characteristics vary with motor state.
- *Inter-activity relationships:* Ratios between the activities (e.g. making toast, putting on jacket) were analyzed by computing the ratio of tremor amplitude (using RMS), with respect to the baseline activity opening door. This enabled characterization of activity dependent modulation of tremor and activity scaling.

- *Body-location analysis:* Features were analyzed per sensor location (e.g., upper arm, lower arm, head) to quantify spatial variation in tremor amplitude and frequency.
- *Distribution coverage:* The similarity between simulated and real tremor was evaluated by measuring how well the simulated feature distributions cover the corresponding distributions observed in the Parkinson’s data.

Comparative analysis

The analysis was performed on three data sources:

- Parkinson patient data (PD)
- Control data (clean baseline)
- Simulated tremor data (SIM)

3.5.4 Activity-dependent tremor behavior

A clear dependency between tremor amplitude and activity type was observed in the PD dataset. This supports the use of an activity-dependent scaling factor in the simulation model,

$$A_{\text{scaled}} = A \cdot \beta_{\text{activity}}, \quad (3.7)$$

where A represents the base tremor amplitude and β_{activity} controls the variation in tremor amplitude across activities.

It further revealed additional complexity beyond the initial simulation model, as different tremor amplitude ranges corresponding to different tremor severities exhibited different activity-dependent behavior. Consequently, the activity-dependent scaling factor β_{activity} was extended to depend on both activity type and tremor severity score.

The activity-dependent scaling factor β_{activity} was defined as a function of both activity type and tremor severity score. The values were determined based on a combination of empirical observations from *Dataset 2* and physiological considerations from the literature [TODO:source]. Activities from the Parkinson dataset (calibration, key, cardigan, and toast) were mapped to corresponding activities in the MHEALTH dataset, and the relative tremor amplitudes observed in the analysis were used to guide the scaling factors.

Table 3.5.5 summarizes the final values used for β_{activity} .

Table 3.5.5: Activity-dependent scaling factors β_{activity} for each tremor severity score.

Activity	Score 1	Score 2	Score 3	Score 4
Standing still	0.80	0.89	0.78	0.72
Sitting and relaxing	1.10	1.10	1.15	1.20
Lying down	1.05	1.05	1.08	1.10
Walking	0.80	0.89	0.78	0.72
Climbing stairs	0.90	0.94	0.89	0.86
Waist bends forward	1.00	1.00	1.00	1.00
Frontal elevation of arms	0.80	0.82	0.90	0.72
Knees bending (crouching)	1.00	1.00	1.00	1.00
Cycling	0.70	0.70	0.70	0.65
Jogging	1.00	1.00	1.00	1.00
Running	1.00	1.00	1.00	1.00
Jump front & back	1.00	1.00	1.00	1.00

3.5.5 Body-location-dependent tremor behavior

The empirical analysis revealed clear spatial differences in tremor amplitude across sensor locations. Lower-arm sensors generally exhibited the highest tremor amplitudes, while upper-arm sensors showed lower amplitudes and the head-mounted sensor exhibited substantially weaker tremor activity.

For example, the ratio between lower-arm and upper-arm tremor amplitudes was approximately

$$\frac{RMS_{\text{lower arm}}}{RMS_{\text{upper arm}}} \approx 1.5. \quad (3.8)$$

These observations support the use of the body-location-dependent scaling factor,

$$A_{\text{scaled}} = A \cdot \beta_{\text{activity}} \cdot \gamma_{\text{location}}, \quad (3.9)$$

where γ_{location} controls the variation in tremor amplitude across body locations.

The empirical analysis further provided data-driven refinement of the location scaling factors used in the simulation, allowing the simulated tremor amplitudes to better reflect the spatial tremor patterns observed in the Parkinson dataset.

3.5.6 Frequency distribution analysis

The dominant tremor frequency in the PD dataset was primarily observed in the range 3–5 Hz, with additional occurrences at higher frequencies extending up to approximately 7–10 Hz.

These observations support the chosen simulation frequency range and indicate that tremor signals may also exhibit higher-frequency components beyond the primary range.

3.5.7 Amplitude analysis

The analysis revealed that simulated tremor amplitudes were generally higher than those observed in the PD dataset. This is reflected in the global statistics, where the median accelerometer RMS in simulated data exceeded that of the PD data.

However, the PD dataset does not include clinical tremor severity annotations and the distribution of tremor severity across subjects is therefore unknown. As a result, direct comparison of global amplitude levels between simulated and real data cannot be directly compared, as the simulated data contains an equal number of segments from each tremor severity level, whereas the PD dataset may be dominated by milder tremor cases. As the simulation model is not intended to reproduce the exact distribution of amplitudes across a population, but rather to generate realistic tremor signals corresponding to different severity levels, this is not considered an issue.

The analysis also showed that the mean of the highest-amplitude windows (top 50) was approximately 4.95 m/s^2 . This aligns closely with the upper severity level (score 4) in the simulation model ($4.65\text{--}5.10 \text{ m/s}^2$), supporting that the selected amplitude range is consistent with the strongest tremor observed in the dataset.

Data-driven validation of the gyroscope-to-acceleration scaling factor (k_g)

In addition to the literature-based estimates, the gyroscope-to-acceleration scaling factor k_g was also estimated directly from *Dataset 2* to ensure consistency with the signal representation used in the simulation pipeline.

For each signal window, k_g was computed as the ratio between the gyroscope and accelerometer RMS values,

$$k_g = \frac{RMS_{\text{gyro}}}{RMS_{\text{acc}}}. \quad (3.10)$$

The gyroscope measurements in the dataset are expressed in radians per second, while the accelerometer measurements are given in m/s^2 . To enable comparison with the literature values, which are reported in degrees per second, the estimated k_g values were converted to units of deg/s per m/s^2 .

Across all activities and subjects, the resulting distribution of k_g values exhibited a relatively narrow spread. The median value was approximately

$$k_g \approx 19 \text{ deg/s per m/s}^2,$$

with an interquartile range of approximately

$$k_g \approx 16\text{--}22 \text{ deg/s per m/s}^2.$$

These values closely match the range obtained from the literature, confirming that the literature-based estimates provide a realistic representation of the relationship between accelerometer and gyroscope tremor amplitudes.

Based on this agreement, the literature-derived values were retained in the simulation model, while the dataset analysis serves as empirical validation of their plausibility.

3.5.8 Impact on downstream classification

- TODO: provide the performance results

To assess the practical usefulness of the refined simulation, models were trained on clean data and on data augmented with simulated tremor, and evaluated on real Parkinson patient data.

The results showed improved performance when simulated tremor was included during training, indicating that the simulation captures relevant characteristics of real tremor. This supports the use of simulated tremor as a data augmentation strategy for activity recognition in the presence of pathological motion disturbances.

3.6 HAR model

3.6.1 Overall fusion architecture

Based on the feature extraction and CNN architectures described in Section 2.3 and Section 2.4, a feature-level fusion HAR model was developed using dedicated 1D CNN feature extractors for each sensor modality.

The proposed HAR system follows a feature-level fusion architecture in which each sensor stream is processed independently through a dedicated 1D CNN feature extractor. Prior to feature extraction, each sensor signal is preprocessed using the pipeline described in Section 3.2.

For each sensor modality, the corresponding CNN extracts a compact latent feature representation in the form of a 128-dimensional embedding vector. The extracted embeddings from all sensor streams are subsequently concatenated into a joint feature representation.

The concatenated feature vector is then used as input to a fusion classifier, which performs the final activity classification. This architecture enables sensor-specific feature learning while allowing information from multiple sensor modalities to be combined at the feature level.

An overview of the proposed HAR framework is illustrated in Figure 3.6.1.

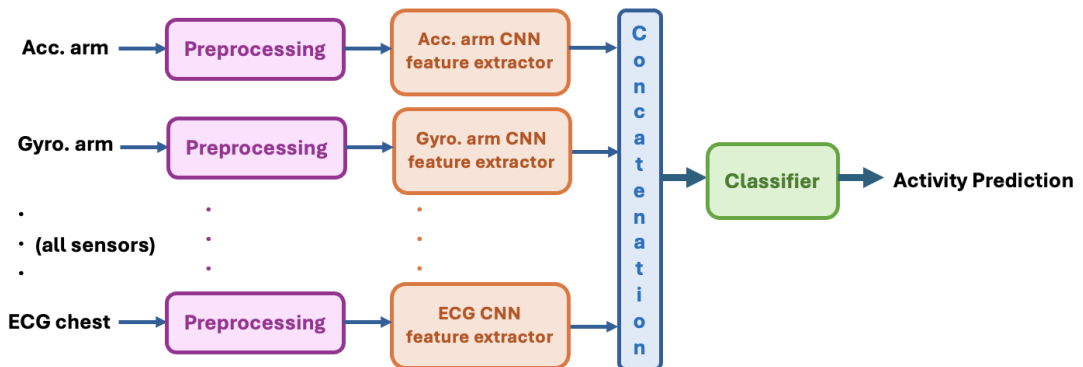


Figure 3.6.1: Overview of the feature-level fusion HAR framework. Each sensor stream is preprocessed independently and processed through a dedicated CNN feature extractor. The extracted embeddings are concatenated and used as input to the final classifier for activity prediction.

3.6.2 CNN architecture

All sensor modalities were processed using identical 1D CNN feature extractors, differing only in the number of input channels depending on the sensor modality. Each feature extractor consisted of two convolutional blocks followed by an embedding layer and a classification head.

The convolutional blocks each contained an 1D convolutional layer with 128 filters and kernel size 5, followed by batch normalization and a ReLU activation function. Dropout regularization was applied after each convolutional block with dropout rates of 0.3.

An adaptive average pooling layer with output size 1 was applied after the second convolutional block in order to reduce the temporal dimension and produce a fixed-length feature representation. In contrast to conventional pooling operations with manually predefined kernel sizes and strides, adaptive pooling automatically adjusts the pooling parameters based on the input size and desired output dimension. This enables the network to produce consistent embedding dimensions even when the input sequence length changes due to different sampling frequencies or preprocessing configurations.

The resulting feature maps were flattened and passed through a fully connected embedding layer producing a 128-dimensional latent feature representation. These embeddings were subsequently extracted and used for feature-level fusion.

During standalone training of the feature extractors, a classification head consisting of dropout regularization and a fully connected output layer was used for activity classification. The classification head was only used during feature extractor training and evaluation, while the extracted embeddings were used as input to the multisensor fusion framework.

The architecture consisted of the following layers:

Convolutional block 1

- *Conv1D layer (128 filters, kernel size 5, padding 2)*
- *Batch normalization*
- *ReLU activation*
- *Dropout (0.3)*

Convolutional block 2

- *Conv1D layer (128 filters, kernel size 5, padding 2)*
- *Batch normalization*
- *ReLU activation*
- *Adaptive average pooling (output size 1)*
- *Dropout (0.3)*

Embedding extraction

- *Flatten layer*

- *Fully connected embedding layer (128-dimensional embedding)*

Classification head

- *Dropout (0.5)*
- *Fully connected classification layer*

The architecture is illustrated in Figure 3.6.2.

One feature extractor total_params: 102,668, trainable: 102,668

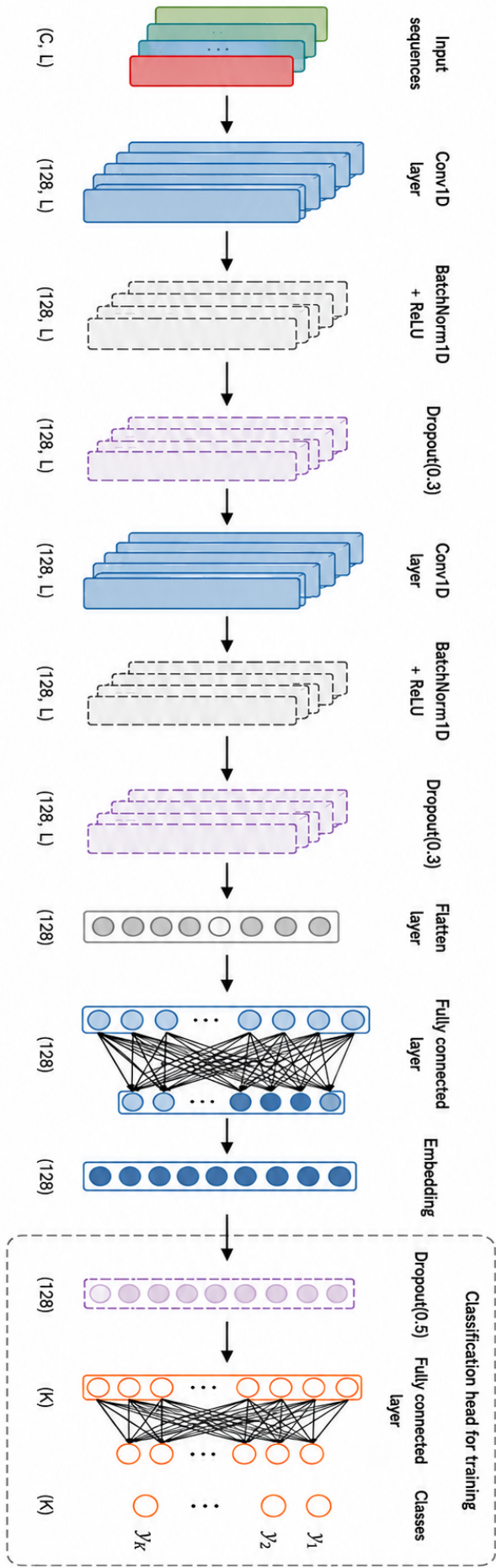


Figure 3.6.2: Architecture of the proposed 1D CNN feature extractor. Here, C denotes the number of input channels, L denotes the input sequence length, and K denotes the number of activity classes. (AI generated figure)

3.7 Fusion Evaluation framework

3.8 Evaluation metrics

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APPENDICES

A - GITHUB REPOSITORY

All code and latex-files used in this document are included in the Github repository linked below. Further explanations are given in the readme-file.

Github repository link

- https://github.com/ninasalvesen/thesis_latex_template

B - SIDENOTE STATISTICS

B1 - Some random table

Remember to only include one thing per page in the appendices.

Statistic	One	Two
Count	387317	283960
Mean	130.66	134.18
Std	248.09	230.32
Q1	31.00	21.00
Median	67.00	63.00
Q3	142.00	159.00
Min	0.00	0.00
Max	14519.00	14253.00

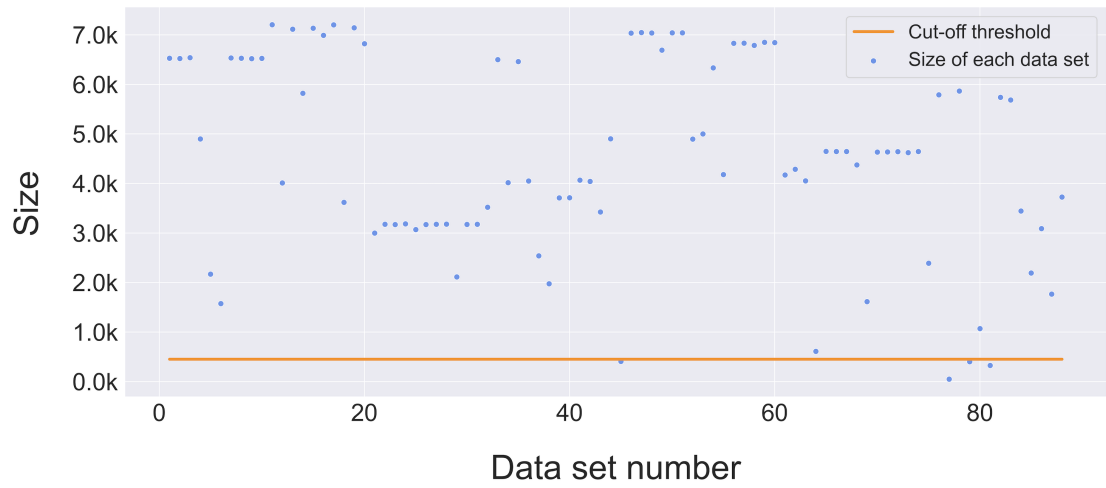
Table B.1: Table of statistics on some sidenote data.

B2 - Some other random table

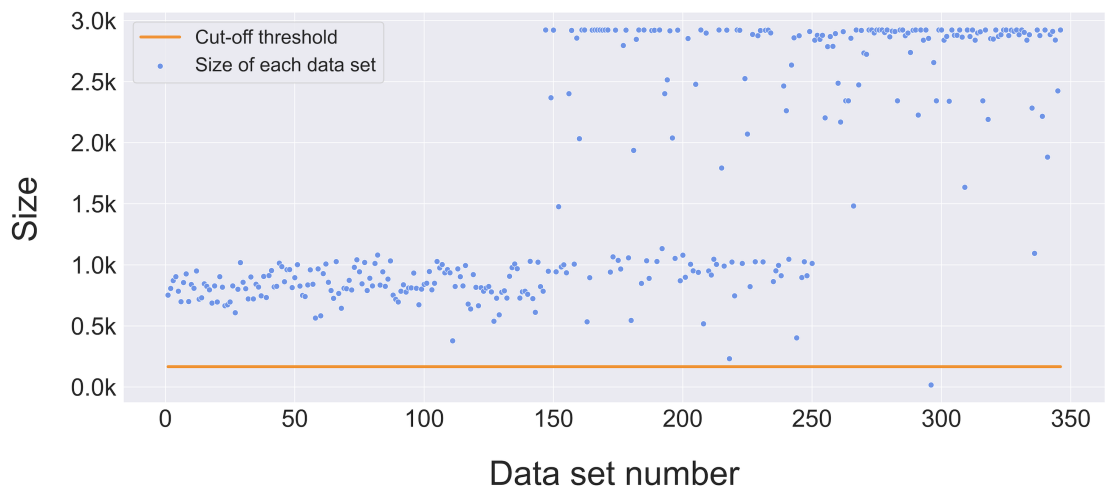
Statistic	Three	Four
Count	387317	283960
Mean	130.66	134.18
Std	248.09	230.32
Q1	31.00	21.00
Median	67.00	63.00
Q3	142.00	159.00
Min	0.00	0.00
Max	14519.00	14253.00

Table B.2: Table of statistics on some other sidenote data.

B3 - Some random figure



(a) Data set sizes for data X.



(b) Data set sizes for data Y.

Figure B.1: The figures show the data set sizes of X and Y and the proposed cut-off threshold at 10 % of the mean set size.